Nimra Nayyar

Software Engineer | Applied AI Systems
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Education

Clemson University — Aug. 2023 – Dec. 2026
Bachelor's of Science in Computer Science, Minor in Artificial Intelligence | GPA: 4.0/4.0 — *Clemson, SC*
- Outreach Chair, Clemson AI Club (200+ members) | SWE Mentor, Clemson Forge (30+ mentees) | School of Computing Ambassador

Technical Skills

Languages: Python, C/C++, Java, JavaScript/TypeScript, SQL, Bash
Systems & Backend: Git, React, Next.js, FastAPI, Flask, REST APIs, Model Context Protocol (MCP), Docker, Linux, CI/CD, Slurm, HPC (Palmetto 2), CUDA
AI/ML: PyTorch, TensorFlow, Transformers, RAG, NLP, Deep Learning, vLLM, NVIDIA NeMo/NIM, YOLO, Hugging Face
Cloud & Data: PostgreSQL, Milvus, pgvector, Cloudflare R2, Supabase

Experience

AI/ML Intern — Jan. 2026 – Present
NVIDIA — *Remote*
- Co-developed **VANTAGE-Bench**, an open benchmark for evaluating vision-language models for Physical AI, comprising **35,027** expert annotations across **3,346** multimodal assets and **four reasoning pillars**; **NeurIPS 2026** submission under review and featured in NVIDIA's GTC Taipei Cosmos-3 keynote.
- Extended VLMEvalKit with custom benchmark datasets, evaluation adapters, and automated scoring infrastructure, enabling reproducible evaluation across frontier vision-language models including **GPT-5.6**, Gemini, and Cosmos-3.
- Built and maintain a production benchmark platform with secure submission APIs, cloud-native evaluation services, reviewer dashboards, and an automated public leaderboard using Next.js, Supabase, Cloudflare R2, and Hugging Face; **live** today, actively accepting submissions, and drawing **3,395+ downloads/month**.
- Designed scalable data engineering pipelines for dataset curation, preprocessing, pseudo-label generation, and validation across two of the benchmark's core tasks, enabling reproducible evaluation of large-scale multimodal models.

Software Engineering Intern (AI Systems) — May 2026 – Aug. 2026
TD SYNNEX — *Greenville, SC*
- Led a 4-person team building the foundational AI automation platform for Cisco sales operations at TD SYNNEX, North America's largest Cisco distributor business: a production multi-agent email-triage system (spam filtering, thread summarization/dedup, intent classification, orchestrator-coordinated agents, MCP-based tool calling) processing **600,000 emails annually**.
- Owned the full SDLC (BRD, design, spec-driven implementation, testing, deployment) targeting **85%+** auto-triage without manual intervention, cutting p95 time-to-ingestion from ~**60 minutes to under 5**, and projecting **$3M** in annual savings and **18.5 FTE** of reclaimed capacity.

AI Research Engineer — Jan. 2026 – May 2026
Clemson Watt AI Program | MUSC Medical Center — *Clemson, SC*
- Engineered a computer vision pipeline extracting structured clinical data from scanned EHR audiograms with MUSC, on Clemson's Palmetto 2 HPC cluster: benchmarked **6 YOLO architectures** (**0.887 recall**) feeding a two-stage pipeline achieving **91% classification accuracy** with sub-decibel error across **20+** categories.
- Built a multimodal video-understanding pipeline (transcription, speaker diarization, multi-condition VLM analysis) and a name-entity-resolution system for social media content, validated via an information-gain evaluation across 154 videos.

Software Engineering Intern (AI Systems) — May 2025 – Aug. 2025
TD SYNNEX — *Greenville, SC*
- Led a small (2–4 person) team building **"Greenville Knowledge Hub"** on NVIDIA's RAG Blueprint (FastAPI, Next.js, Docker): **24+** REST endpoints, dual vector-store architecture (Milvus/pgvector) across **5** knowledge bases.
- Implemented hybrid retrieval (**40**-candidate rerank-to-top-k, LLM-classified fallback to auto-generated SQL) behind fail-closed NeMo Guardrails and JWT/RBAC-secured multi-tenant access; delivered a live demo to **3,000+ executives** at TD SYNNEX Inspire/Vendor Summit.

Projects

PromptPact | *Python, Typer, Pydantic, MCP, OpenAI, Anthropic, Ollama* — 2026
- **Open-source LLM developer infrastructure tool** for prompt regression testing using deterministic baseline comparisons across prompt and model updates.
- Designed a provider-agnostic execution engine with pluggable model backends and configurable target vs. judge model separation for automated LLM evaluation.
- Integrated an MCP server exposing run, diff, and list_cases tools for AI coding agents while enforcing human-controlled baseline creation.

MuseBoard | *Python, Gemini, FAISS, Sentence Transformers, SQLite, Notion API* — 2026
- Built an end-to-end pipeline that converts Pinterest boards into **structured marketing intelligence** using vision-language models, semantic embeddings, clustering, and Notion integration.
- Engineered a modular six-stage architecture with schema validation, local vector search (FAISS), retryable API orchestration, and caching for reliable, reproducible execution.